

# Special Case Integration

Numberbender | WORKSHEET



Name: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

Find the antiderivative of each. Show all work.

## AB Calculus Worksheet #30

Find the antiderivative of each of the following, given:  $\int a^x dx = \frac{1}{\ln a} a^x + C$

### Examples:

|    |                                                                                                                                                                                                                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | $\int 8^x dx = \boxed{\frac{1}{\ln 8}} 8^x + C = \boxed{\frac{1}{\ln 8}} \bullet 8^x + C$ $\frac{d}{dx}(8^x) = \ln 8 \bullet 8^x \Rightarrow ck : 8^x = \boxed{\ln 8} \bullet 8^x \Rightarrow \boxed{\ln 8} = \frac{1}{\ln 8}$                                                                                                                                               |
| 2. | $\int_0^3 3^{2x} dx = \boxed{\frac{1}{2 \ln 3}} \bullet 3^{2x} \Big _0^3 = \frac{1}{2 \ln 3} (3^{2(3)} - 3^{2(0)}) = \frac{1}{2 \ln 3} (3^6 - 1) = \boxed{\frac{728}{2 \ln 3}}$ $\frac{d}{dx}(3^{2x}) = 2 \ln 3 \bullet 3^{2x} \Rightarrow ck : 3^{2x} = \boxed{\frac{1}{2 \ln 3}} \bullet 2 \ln 3 \bullet 3^{2x} \Rightarrow \boxed{\frac{1}{2 \ln 3}} = \frac{1}{2 \ln 3}$ |

### Problems:

|                                                  |                                          |                                       |
|--------------------------------------------------|------------------------------------------|---------------------------------------|
| 1. $\int 1.3^x dx$                               | 2. $\int_{-2}^0 5^{-\theta} d\theta$     | 3. $\int_1^{\sqrt{2}} 2^{x^2} x dx$   |
| 4. $\int_0^{\frac{\pi}{2}} 7^{\cos t} \sin t dt$ | 5. $\int 2^x dx$                         | 6. $\int 5^x dx$                      |
| 7. $\int_0^1 2^{-\theta} d\theta$                | 8. $\int e^{3x} dx$                      | 9. $\int_1^2 \frac{2^{\ln x}}{x} dx$  |
| 10. $\int_{-2}^2 5 dx$                           | 11. $\int_0^{\frac{\pi}{4}} \cos x dx$   | 12. $\int_0^1 (8s^3 - 12s^2 + 5) ds$  |
| 13. $\int_1^{27} y^{-\frac{4}{3}} dy$            | 14. $\int_0^{\frac{\pi}{2}} \sec^2 x dx$ | 15. $\int_0^1 \frac{36}{(2x+1)^3} dx$ |



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Labanan ang agwat.

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### Special Case Integration Worksheet

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# Special Case Integration

Numberbender | ANSWER KEY



Answer key — for instructor use only.

## Answers:

|                                        |                             |                                  |                      |                                    |                                    |
|----------------------------------------|-----------------------------|----------------------------------|----------------------|------------------------------------|------------------------------------|
| 1. $\frac{1}{\ln 1.3} \cdot 1.3^x + C$ | 2. $\frac{24}{\ln 5}$       | 3. $\frac{1}{\ln 2}$             | 4. $\frac{6}{\ln 7}$ | 5. $\frac{1}{\ln 2} \cdot 2^x + C$ | 6. $\frac{1}{\ln 5} \cdot 5^x + C$ |
| 7. $\frac{1}{2 \ln 2}$                 | 8. $\frac{1}{3} e^{3x} + C$ | 9. $\frac{2^{\ln 2} - 1}{\ln 2}$ | 10. 20               | 11. $\frac{\sqrt{2}}{2}$           | 12. 3                              |
| 13. 2                                  | 14. $\sqrt{3}$              | 15. 8                            |                      |                                    |                                    |

